DEMOSAT-1 SW user manual

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1 Introduction

1.1 Purpose

This manual describes the on-board software (SCSW) of DEMOSAT-1 for the engineers who operate the spacecraft. It explains what each application does, which telecommands it accepts and which housekeeping parameters it reports.

The formats of packets and telecommands are defined in the SCSW interface control document. The flight operations procedures use the telecommands described here.

1.2 Scope

The manual covers SCSW version 3.1 and its 6 applications. Ground segment software and the boot loader are out of scope.

1.3 Reference documents

- DEMOSAT-1-SCSW-ICD issue 2.0, SCSW interface control document
- DEMOSAT-1-FOP issue 1.0, flight operations procedures
- DEMOSAT-1-SRDB version 1.2, satellite reference database

2 SCSW overview

2.1 Architecture

The SCSW runs on the on-board computer and is organised in 6 applications, each with its own telecommand dispatcher and housekeeping report: power (PWR), thermal control (TCS), attitude and orbit control (AOC), on-board computer (OBC), communications (COM), payload (PLD).

A scheduler calls every application once per cycle. Telecommands reach an application through its application process identifier (APID).

2.2 Telecommand handling

When a telecommand arrives, the SCSW checks it, sends the acceptance report and passes it to the dispatcher of the application that owns its APID. The dispatcher calls one handler per telecommand; for example PWRC0101 (PWR_HTR_A_ON) is handled by the power application.

Critical telecommands such as PWR_SET_BAT_EOC need an arm and fire sequence before they execute.

2.3 Housekeeping

Every application reports its state in its own housekeeping packets, PUS (3,25): PWR_HK, TCS_HK, AOC_HK, OBC_HK, COM_HK and PLD_HK. Each report can be enabled and disabled by its structure identifier.

3 PUS services

3.1 Services used

Table 3-1 PUS services used

Service	Subtypes	Name	Used for
1	1 to 8	Request verification	Acceptance and execution reports
3	5, 6	Housekeeping	Enable and disable a report
3	25	Housekeeping	Housekeeping parameter report
5	1, 4	Event reporting	Informative and high-severity events
8	1	Function management	Perform a function of an application
17	1	Test	Connection test
20	1, 3	Parameter management	Report and set on-board parameters

3.2 Event reporting

Events are reported in two packets, both sent by the on-board computer application.

Table 3-2 Event packets

Packet	Subtype	Description
EVT_INFO	1	Informative event report
EVT_ALARM	4	High-severity event report

4 Power application (PWR)

4.1 Overview

The power application (PWR) conditions the main bus, charges the battery and controls the two battery heaters. It reports its state every 8 s in the PWR_HK packet.

The main battery voltage BAT_V_MAIN (PWRT0012) and the main battery current BAT_I_MAIN are sampled at 1 Hz and averaged over each reporting period. When PWRT0012 leaves its soft limits, the application raises an event in the EVT_ALARM packet.

The application is commanded with PWRC0101 to switch battery heater A on and with PWRC0112 to switch battery heater A off. The result shows in HTR_A_STAT.

4.2 Telecommands

Table 4-1 PWR telecommands

TC	Mnemonic	Description	Arguments
PWRC0101	PWR_HTR_A_ON	Switch battery heater A on	PWRP0001 (HTR_MODE)
PWRC0112	PWR_HTR_A_OFF	Switch battery heater A off	None
PWRC0123	PWR_SET_BAT_EOC	Set the battery end-of-charge voltage	PWRP0010 (EOC_V)
PWRC2476	PWR_HTR_B_ON	Switch battery heater B on	PWRP0001 (HTR_MODE)
PWRC5141	PWR_LCL_RESET	Reset a tripped latching current limiter	PWRP9190 (LCL_ID)

4.3 Housekeeping

The PWR_HK packet, PUS (3,25), carries the parameters below.

Table 4-2 PWR housekeeping parameters

Parameter	Mnemonic	Unit	Description
PWRT0012	BAT_V_MAIN	V	Main battery voltage
PWRT0281	BAT_I_MAIN	A	Main battery current
PWRT0020	BAT_TEMP_1	degC	Battery temperature, sensor 1
PWRT0324	BAT_TEMP_2	degC	Battery temperature, sensor 2
PWRT0031	HTR_A_STAT		Battery heater A status
PWRT8596	HTR_B_STAT		Battery heater B status
PWRT9647	HTR_A_MODE		Battery heater A control mode
PWRT0045	SA_I_TOTAL	A	Solar array total current
PWRT0694	SA_V_OUT	V	Solar array output voltage
PWRT1392	BUS_V_MAIN	V	Main bus voltage
PWRT5284	BCR_MODE		Battery charge regulator mode
PWRT9077	BAT_EOC_V	V	Battery end-of-charge voltage limit
PWRT8638	LCL_TRIP_CNT		Latching current limiter trip count
PWRT2959	BAT_V	V	Battery voltage

4.4 Operations notes

Note: Before switching a heater on, check that BAT_V_MAIN is above 26 V. In forced mode (PWRP0001 = 2) the heater stays on until PWRC0112 is sent.

Note: Lowering the end-of-charge voltage with PWR_SET_BAT_EOC slows battery ageing. The new limit shows in PWRT9077 at the end of the next charge cycle.

Related procedures:

- FOP-PWR-001 Battery heater activation

5 Thermal control application (TCS)

5.1 Overview

The thermal control application (TCS) keeps every unit within its temperature limits with heater lines and thermistors. It reports its state every 16 s in the TCS_HK packet.

The heater line 1 set point HL1_SETP (TCST9481) and the radiator temperature, panel 1 RAD_TEMP_1 are sampled at 1 Hz and averaged over each reporting period. When TCST9481 leaves its soft limits, the application raises an event in the EVT_ALARM packet.

The application is commanded with TCSC9695 to switch heater line 1 on and with TCSC1330 to switch heater line 1 off. The result shows in HL1_STAT.

5.2 Telecommands

Table 5-1 TCS telecommands

TC	Mnemonic	Description	Arguments
TCSC9695	TCS_HL1_ON	Switch heater line 1 on	TCSP6912 (HL_MODE)
TCSC1330	TCS_HL1_OFF	Switch heater line 1 off	None
TCSC1076	TCS_SET_HL1_SETP	Set the heater line 1 set point	TCSP9614 (SETP_DEGC)
TCSC1484	TCS_MODE_SET	Change the thermal control mode	TCSP2480 (TCS_MODE_REQ)
TCSC0408	TCS_HL2_ON	Switch heater line 2 on	TCSP6912 (HL_MODE)

5.3 Housekeeping

The TCS_HK packet, PUS (3,25), carries the parameters below.

Table 5-2 TCS housekeeping parameters

Parameter	Mnemonic	Unit	Description
TCST6132	HL1_STAT		Heater line 1 status
TCST4051	HL2_STAT		Heater line 2 status
TCST1019	HL3_STAT		Heater line 3 status
TCST7244	HL1_MODE		Heater line 1 control mode
TCST9481	HL1_SETP	degC	Heater line 1 set point
TCST0779	RAD_TEMP_1	degC	Radiator temperature, panel 1
TCST0971	RAD_TEMP_2	degC	Radiator temperature, panel 2
TCST1795	TANK_TEMP	degC	Propellant tank temperature
TCST3723	OBC_BRD_TEMP	degC	On-board computer board temperature
TCST7158	PLD_IF_TEMP	degC	Payload interface temperature
TCST3307	TCS_MODE		Thermal control mode
TCST7930	HL_I_TOTAL	A	Heater lines total current
TCST2520	TH_FAIL_CNT		Thermistor failure count
TCST8951	HL_V	V	Heater line voltage

5.4 Operations notes

Note: Heater line 1 warms the propellant tank. Keep TANK_TEMP between 5 and 45 degC; below 5 degC, check TCST6132 before commanding TCS_HL1_ON.

Note: After a thermistor failure, TH_FAIL_CNT increases and the affected line falls back to forced mode.

Related procedures:

- FOP-TCS-001 Heater line 1 activation

6 Attitude and orbit control application (AOC)

6.1 Overview

The attitude and orbit control application (AOC) points the spacecraft with gyros, sun sensors, a star tracker and three reaction wheels. It reports its state every 4 s in the AOC_HK packet.

The gyro rate, X axis GYR_RATE_X (AOCT2611) and the gyro rate, Y axis GYR_RATE_Y are sampled at 1 Hz and averaged over each reporting period. When AOCT2611 leaves its soft limits, the application raises an event in the EVT_ALARM packet.

The application is commanded with AOCC1067 to change the attitude control mode and with AOCC6804 to switch the star tracker on. The result shows in AOC_MODE.

6.2 Telecommands

Table 6-1 AOC telecommands

TC	Mnemonic	Description	Arguments
AOCC1067	AOC_MODE_SET	Change the attitude control mode	AOCP9017 (AOC_MODE_REQ)
AOCC6804	AOC_STR_ON	Switch the star tracker on	None
AOCC9208	AOC_STR_OFF	Switch the star tracker off	None
AOCC7889	AOC_RW_SPEED_SET	Set a reaction wheel speed	AOCP7625 (RW_ID), AOCP5945 (RW_SPEED_REQ)
AOCC6352	AOC_GYR_RESET	Reset the gyro bias estimate	None

6.3 Housekeeping

The AOC_HK packet, PUS (3,25), carries the parameters below.

Table 6-2 AOC housekeeping parameters

Parameter	Mnemonic	Unit	Description
AOCT2478	AOC_MODE		Attitude control mode
AOCT2611	GYR_RATE_X	deg/s	Gyro rate, X axis
AOCT6114	GYR_RATE_Y	deg/s	Gyro rate, Y axis
AOCT5392	GYR_RATE_Z	deg/s	Gyro rate, Z axis
AOCT0240	SUN_ANGLE	deg	Sun sensor angle
AOCT8321	RW1_SPEED	rpm	Reaction wheel 1 speed
AOCT4403	RW2_SPEED	rpm	Reaction wheel 2 speed
AOCT9618	RW3_SPEED	rpm	Reaction wheel 3 speed
AOCT2376	MTQ_X_DIPOLE	Am2	Magnetorquer X dipole
AOCT8208	STR_STAT		Star tracker status
AOCT3539	STR_LOCK		Star tracker lock
AOCT0932	ATT_ERR	deg	Attitude error
AOCT8473	RW1_TEMP	degC	Reaction wheel 1 temperature

6.4 Operations notes

Note: Do not command AOC_MODE_SET to fine pointing before STR_LOCK reports LOCKED.

Note: Wheel speeds above 5000 rpm in AOCT8321 trigger a momentum unload with the magnetorquers.

Related procedures:

- FOP-AOC-001 Transition to nadir pointing

7 On-board computer application (OBC)

7.1 Overview

The on-board computer application (OBC) runs the software, keeps on-board time and handles housekeeping and events. It reports its state every 8 s in the OBC_HK packet.

The processor load CPU_LOAD (OBCT3700) and the free memory MEM_FREE are sampled at 1 Hz and averaged over each reporting period. When OBCT3700 leaves its soft limits, the application raises an event in the EVT_ALARM packet.

The application is commanded with OBCC2858 to enable a housekeeping report and with OBCC8719 to disable a housekeeping report. The result shows in HK_ENA_CNT.

7.2 Telecommands

Table 7-1 OBC telecommands

TC	Mnemonic	Description	Arguments
OBCC2858	OBC_HK_ENABLE	Enable a housekeeping report	OBCP4382 (SID)
OBCC8719	OBC_HK_DISABLE	Disable a housekeeping report	OBCP4382 (SID)
OBCC9182	OBC_CONN_TEST	Perform a connection test	None
OBCC2725	OBC_PAR_REPORT	Report on-board parameter values	OBCP6968 (PAR_ID)
OBCC1080	OBC_PAR_SET	Set an on-board parameter value	OBCP6968 (PAR_ID), OBCP8975 (PAR_VAL)

7.3 Housekeeping

The OBC_HK packet, PUS (3,25), carries the parameters below.

Table 7-2 OBC housekeeping parameters

Parameter	Mnemonic	Unit	Description
OBCT3700	CPU_LOAD	%	Processor load
OBCT9283	MEM_FREE	kB	Free memory
OBCT4639	WDG_STAT		Watchdog status
OBCT0518	RESET_CNT		Reset count
OBCT4211	EDAC_ERR_CNT		Memory error count
OBCT2880	OBT_DRIFT	ms	On-board time drift
OBCT6792	SW_MODE		Software mode
OBCT5611	TC_ACC_CNT		Accepted telecommand count
OBCT9701	TC_REJ_CNT		Rejected telecommand count
OBCT4459	EVT_LAST_ID		Last event identifier
OBCT8109	EVT_CNT		Event count
OBCT1310	RAIL_V_5V	V	5 V rail voltage
OBCT9444	HK_ENA_CNT		Enabled housekeeping report count

7.4 Operations notes

Note: After a reset, RESET_CNT increases and every housekeeping report except the OBC_HK packet is disabled until OBCC2858 is sent.

Note: Watch OBCT9701 during a pass; a rising count means telecommands fail acceptance.

Related procedures:

- FOP-OBC-001 Housekeeping report recovery

8 Communications application (COM)

8.1 Overview

The communications application (COM) runs the S-band receiver and transmitter and selects the antenna. It reports its state every 8 s in the COM_HK packet.

The receiver signal level RX_AGC (COMT7144) and the receiver signal-to-noise ratio RX_SNR are sampled at 1 Hz and averaged over each reporting period. When COMT9236 leaves its soft limits, the application raises an event in the EVT_ALARM packet.

The application is commanded with COMC3469 to switch the transmitter on and with COMC0655 to switch the transmitter off. The result shows in TX_STAT.

8.2 Telecommands

Table 8-1 COM telecommands

TC	Mnemonic	Description	Arguments
COMC3469	COM_TX_ON	Switch the transmitter on	COMP9910 (TX_RATE_SEL)
COMC0655	COM_TX_OFF	Switch the transmitter off	None
COMC1263	COM_XPD_MODE_SET	Change the transponder mode	COMP1462 (XPD_MODE_REQ)
COMC5525	COM_ANT_SELECT	Select the active antenna	COMP9173 (ANT_ID)
COMC0709	COM_RX_RESET	Reset the receiver	None

8.3 Housekeeping

The COM_HK packet, PUS (3,25), carries the parameters below.

Table 8-2 COM housekeeping parameters

Parameter	Mnemonic	Unit	Description
COMT5338	RX_LOCK		Receiver lock
COMT7144	RX_AGC	dBm	Receiver signal level
COMT9701	RX_SNR	dB	Receiver signal-to-noise ratio
COMT4549	TX_STAT		Transmitter status
COMT9236	TX_PWR	W	Transmitter output power
COMT7650	TX_TEMP	degC	Transmitter temperature
COMT6242	TX_RATE	kbps	Downlink rate
COMT1448	XPD_MODE		Transponder mode
COMT7974	RX_FREQ_ERR	Hz	Receiver frequency offset

Parameter	Mnemonic	Unit	Description
COMT3984	ANT_SEL		Antenna selection
COMT4629	TX_I	A	Transmitter current
COMT9823	RX_TEMP	degC	Receiver temperature
COMT8772	FRAME_CNT		Transmitted frame count

8.4 Operations notes

Note: Switch the transmitter off with COM_TX_OFF at the end of every pass; TX_TEMP rises by about 10 degC per hour of transmission.

Related procedures:

- FOP-COM-001 Downlink start

9 Payload application (PLD)

9.1 Overview

The payload application (PLD) operates the camera, its cooler and the image store. It reports its state every 16 s in the PLD_HK packet.

The focal plane temperature FPA_TEMP (PLDT0056) and the camera current CAM_I are sampled at 1 Hz and averaged over each reporting period. When PLDT0056 leaves its soft limits, the application raises an event in the EVT_ALARM packet.

The application is commanded with PLDC1133 to switch the camera on and with PLDC4748 to switch the camera off. The result shows in CAM_STAT.

9.2 Telecommands

Table 9-1 PLD telecommands

TC	Mnemonic	Description	Arguments
PLDC1133	PLD_CAM_ON	Switch the camera on	PLDP3324 (CAM_MODE_REQ)
PLDC4748	PLD_CAM_OFF	Switch the camera off	None
PLDC9313	PLD_IMG_TAKE	Take a series of images	PLDP3136 (EXP_MS), PLDP7412 (IMG_N)
PLDC8017	PLD_COOLER_ON	Switch the focal plane cooler on	None
PLDC2062	PLD_STORE_CLEAR	Clear the image store	None

9.3 Housekeeping

The PLD_HK packet, PUS (3,25), carries the parameters below.

Table 9-2 PLD housekeeping parameters

Parameter	Mnemonic	Unit	Description
PLDT8785	CAM_STAT		Camera status
PLDT3063	CAM_MODE		Camera mode
PLDT0056	FPA_TEMP	degC	Focal plane temperature
PLDT2237	CAM_I	A	Camera current
PLDT7193	CAM_V	V	Camera supply voltage

Parameter	Mnemonic	Unit	Description
PLDT8768	IMG_CNT		Image count
PLDT8949	EXP_TIME	ms	Exposure time
PLDT7797	COOLER_STAT		Cooler status
PLDT1679	COOLER_PWR	W	Cooler power
PLDT6795	STORE_FILL	%	Image store fill level
PLDT7871	OPT_TEMP	degC	Optics temperature
PLDT7699	SHUTTER_STAT		Shutter status
PLDT9158	CMP_RATIO		Compression ratio

9.4 Operations notes

Note: Images are only usable once FPA_TEMP is below -15 degC; allow ten minutes after PLDC8017.

Note: Clear the image store with PLD_STORE_CLEAR only after the ground confirms the downlink of every image.

Related procedures:

- FOP-PLD-001 Imaging session

A Acronyms

Acronym	Meaning
APID	Application process identifier
FOP	Flight operations procedure
HK	Housekeeping
ICD	Interface control document
PUS	Packet utilisation standard
SCSW	Spacecraft software
SPID	Packet identifier in the reference database
SRDB	Satellite reference database
SUM	Software user manual
TC	Telecommand
TM	Telemetry